CONCENTRATION & EXIT RISK ASSESSMENT

# AWS exit risk review: demo organisation

Synthetic data for demonstration. Not a real organisation.

SCOPE
name contains 'prod'

ORGANIZATIONAL UNIT
Root

ACCOUNTS IN SCOPE
40 of 49

SPEND PERIOD
Oct 2025 to Sep 2026 (12 months, amortised)

SCANNED
Oct 21, 2026, 11:50 AM

MANAGEMENT ACCOUNT
111111111111

RESOURCE INVENTORY
Included (deep scan)

EXIT TARGET ASSUMED
Microsoft Azure

# 1 Executive summary

BOTTOM LINE

## Exit risk is Moderate. Most workloads are portable, but the size of the estate makes an exit a large programme. Cost and time are not yet estimated for your organisation: answer the delivery questions under Exit model.

Scores 39/100 by spend and 28/100 by number of accounts. Risk is concentrated: 6 accounts (36% of spend), mostly on Redshift, DynamoDB and Lambda, carry most of it, while 34 of 40 accounts (85%) are broadly portable.

**Cost and time are provisional.** They use default answers, not your organisation's. Answer the delivery questions (CI/CD, Terraform, team size, day rate) to estimate them for you.

### WHAT WE FOUND

- **27% of spend ($2.7m a year)** runs on 19 bespoke, AWS-only services that must be re-engineered: Redshift $746k a year (1 account), DynamoDB $530k (6) and Lambda $391k (10); 16 others $1.0m.
- **34 of 40 accounts are portable** (Low or Moderate risk). The 6 High or Critical accounts hold 36% of spend.
- **3 accounts set the pace:** data-platform-prod, cards-processing-prod and contact-centre-prod carry 57% of the spend at risk (spend x risk).
- **Spend grew 42%** from Oct 2025 to Sep 2026; High/Critical spend grew 38%, so lock-in is growing more slowly than the estate. Biggest monthly rises: EC2 +$62k, RDS / Aurora +$38k and Redshift +$23k.
- **The exit does not fit the 24-month deadline:** it takes roughly 2.5 to 5 years on default delivery answers. Start earlier, add capacity, or agree a longer transition.
- **Commitments outlast the exit:** $10k to $100k of Savings Plans and reservations would still be owed when it finishes; the last commitment ends Sep 2029.
- **90% of the spend of 5 critical accounts** runs in Europe (London), so one regional event takes down most of it at once. Channels Card payments, Core banking, Mobile app, Payments and Shared platform share that region: no anti-affinity between them.

### WHAT WE RECOMMEND

1. **Answer the delivery questions under Exit model** (CI/CD, Terraform, team size, day rate). Until then the cost and time figures use default answers, not your organisation's.
2. **Plan the hardest accounts first.** Write and test exit plans for data-platform-prod, cards-processing-prod and contact-centre-prod; they drive most of the cost and the timeline.
3. **Make portability a design criterion** for new work on Redshift, DynamoDB and Lambda: prefer the open-standard alternative in Appendix A unless the business case says otherwise.
4. **Pre-build a stressed-exit fallback** for the 24% of critical spend on AWS-only services: a stand-in process, a manual workaround or a warm copy elsewhere, tested before it is needed.
5. **Match commitment terms to the exit plan:** $318k a year of commitments runs past the fastest exit. Buy new Savings Plans and reservations on terms that end inside the exit window.

**Confidence: Low** Synthetic demo data. Cost and time use default delivery answers, not yours. 1 service not in the catalog, scored as Moderate.

### KEY FIGURES

**OU exit-risk score**

| By spend | By no. of accounts |
|---|---|
| 39/100 | 28/100 |
| Moderate | Moderate |

Risk is concentrated in a few large accounts: most accounts are easier to move than the spend view suggests.




Annual spend in scope
**$9.9m**
$828k / month, amortised

Spend on High/Critical services
**27%**
$2.7m a year

Accounts in scope
**40**
40 with spend · 9 filtered out

High/Critical accounts
**6 of 40**
36% of spend · 34 portable (Low/Moderate)

Bespoke (AWS-only) services
**19**
largest: Redshift, DynamoDB, Lambda

Spend growth
**+41.5%**
Oct 2025 → Sep 2026 · High/Critical +38.5%

Exit cost (one-off)
**$7m to $16m** PROVISIONAL
70 to 180% of a year's AWS spend

Exit window (time to leave AWS)
**2.5 to 5 years** PROVISIONAL
bound by team capacity

Engineering effort
**25 to 60 person-years** PROVISIONAL
team of 15 full-time people

## 2 Where the exit risk sits

### Monthly spend by exit-risk tier

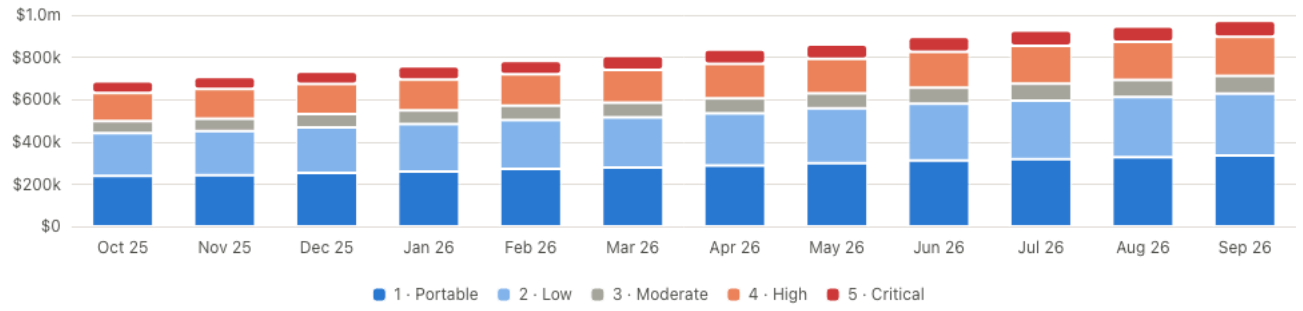
Share of in-scope spend by how hard it is to move.



| Tier | Per month | Share | Services | Accounts |
|---|---|---|---|---|
| 1 · Portable | $288k | 34.8% | 12 | 35 |
| 2 · Low | $245k | 29.6% | 18 | 39 |
| 3 · Moderate | $70k | 8.5% | 11 | 7 |
| 4 · High | $159k | 19.2% | 16 | 14 |
| 5 · Critical | $66k | 7.9% | 3 | 7 |

### Spend trend by tier

Is lock-in growing? Monthly amortised spend.

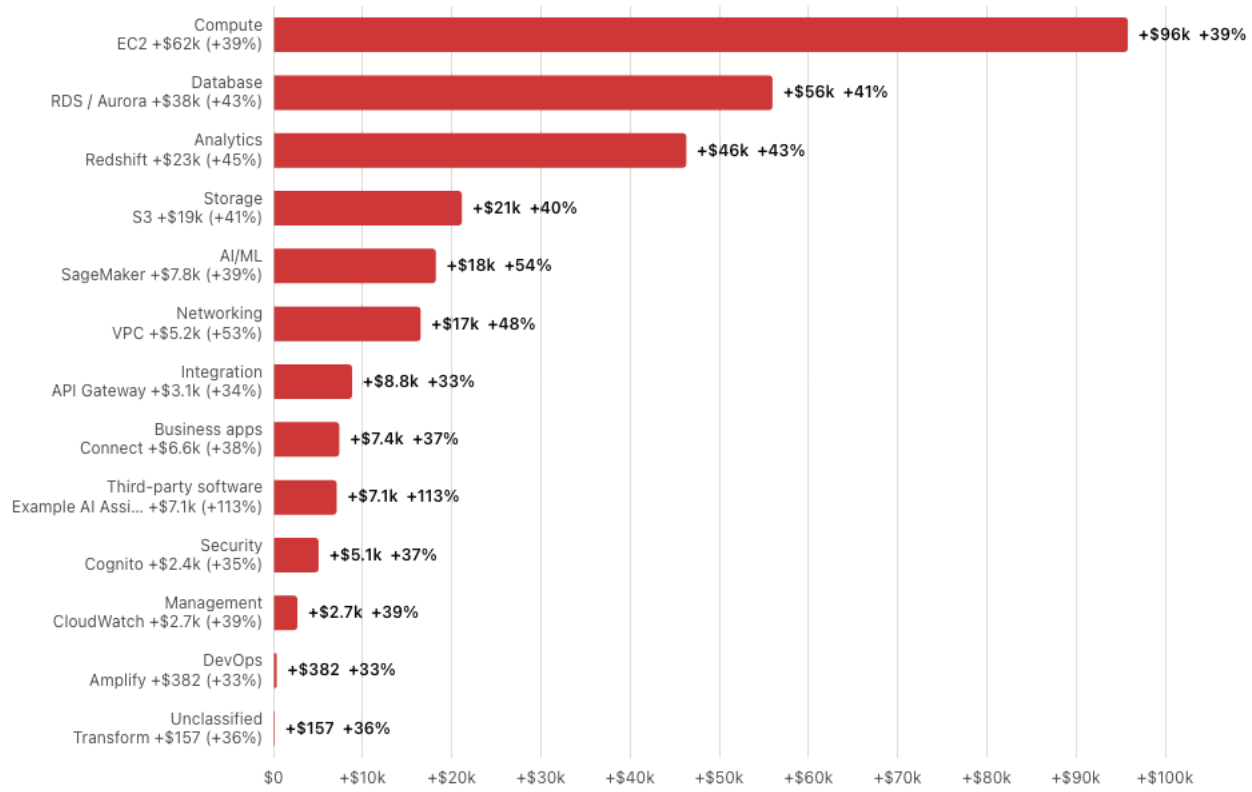



SPEND GROWTH BY PRODUCT TYPE

Oct 2025 to Sep 2026: total spend +41.5% (+46.0% a year); spend on High/Critical services +38.5% (+42.6% a year). Lock-in is growing more slowly than the estate.

### Change in monthly spend

Change in monthly spend from Oct 2025 to Sep 2026, by product type, in dollars and percent. Under each type, the service that moved most. Red = growth, blue = decline.



## Monthly spend by product type

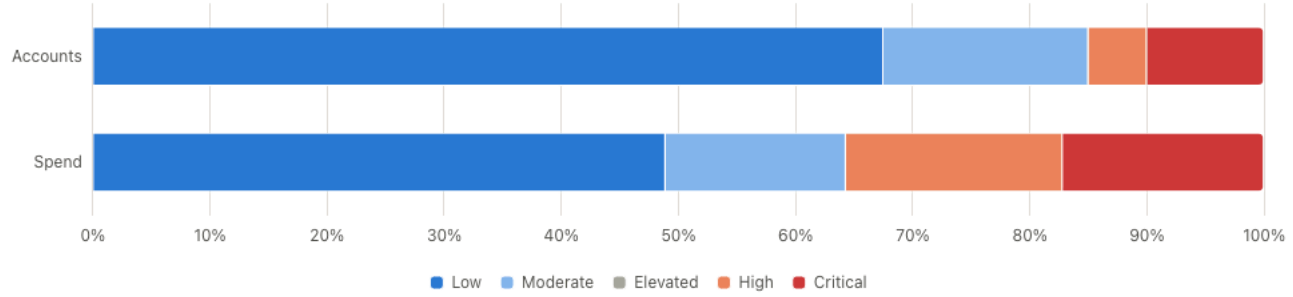
| Product type | Oct 2025 | Sep 2026 | Change | Per year | Lock-in |
|---|---|---|---|---|---|
| Compute<br>EC2, EC2 Other (EBS, NAT, transfer), Lambda, ECS +2 more | $247k | $343k | +38.8% | +43.0% | 11% |
| Database<br>RDS / Aurora, DynamoDB, Neptune, ElastiCache +1 more | $137k | $193k | +40.8% | +45.2% | 28% |
| Analytics<br>Redshift, Glue, EMR, Kinesis Streams +9 more | $108k | $154k | +43.1% | +47.8% | 71% |
| Storage<br>S3, FSx, Backup | $54k | $75k | +39.5% | +43.8% | 0% |
| AI/ML<br>SageMaker, Bedrock, Transcribe, Lex +2 more | $34k | $52k | +54.2% | +60.4% | 10% |
| Networking<br>VPC, Direct Connect, CloudFront, Network Firewall +3 more | $34k | $51k | +47.9% | +53.3% | 0% |
| Integration<br>API Gateway, Step Functions, AppSync, SQS +4 more | $27k | $36k | +32.8% | +36.3% | 45% |
| Business apps<br>Connect, Pinpoint | $20k | $27k | +37.2% | +41.2% | 100% |
| Security<br>Cognito, KMS, Directory Service, ACM / Private CA +2 more | $14k | $19k | +36.8% | +40.7% | 49% |
| Third-party software<br>Marketplace (3rd party) | $6.3k | $13k | +112.5% | +127.5% | 0% |
| Management<br>CloudWatch | $7.0k | $9.7k | +38.6% | +42.8% | 0% |
| DevOps<br>Amplify | $1.1k | $1.5k | +33.4% | +36.9% | 100% |
| Unclassified<br>Transform | $435 | $592 | +36.0% | +39.9% | 0% |
| **Total** | $689k | $974k | **+41.5%** | +46.0% | |

Per year = the first-to-last-month change annualised. Amortised spend: Savings Plans and RIs are spread over the months they cover.

## 3 Accounts: how many are at risk, and how big

### Accounts by exit-risk rating

Share of accounts (top) against share of spend (bottom). Few accounts holding a lot of spend is a concentration to manage, not an estate-wide problem.



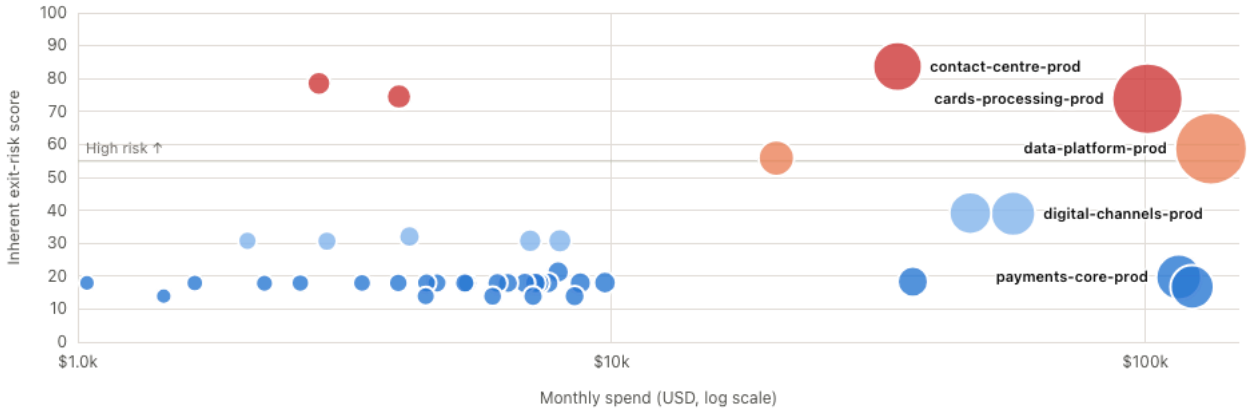

### Portable vs higher-risk accounts

| Rating | Accounts | Share | Per month | Spend share |
|---|---|---|---|---|
| Low | 27 | 67.5% | $405k | 48.9% |
| Moderate | 7 | 17.5% | $128k | 15.4% |
| Elevated | 0 | 0.0% | $0 | 0.0% |
| High | 2 | 5.0% | $153k | 18.5% |
| Critical | 4 | 10.0% | $142k | 17.2% |
| **Portable** (Low + Moderate) | **34** | 85.0% | | |

OU score weighting: By spend. Each account counts in proportion to its spend, so a small high-risk account adds little. It sets bubble sizes and contributions; change it in the bar at the top.

### Inherent risk vs spend

Each bubble is an account. Up = harder to move; right = bigger; bubble size = points it contributes to the OU score. Top-right accounts drive the exit.



## Account ranking by criticality: spend × risk

| # | Account | Per month | Risk | Spend at risk / month | Main lock-in |
|---|---|---|---|---|---|
| 1 | **data-platform-prod**<br>561017993421 · Root / Workloads / Production · Owner: Emma Botha | $133k | High · 59<br>63% High/Crit spend | $78k | Redshift Glue QuickSight |
| 2 | **cards-processing-prod** Critical<br>966773840736 · Card payments · Root / Workloads / Production · Owner: Priya Shah | $101k | Critical · 74<br>77% High/Crit spend | $75k | DynamoDB Lambda Kines |
| 3 | **contact-centre-prod**<br>125285552493 · Contact centre · Root / Workloads / Production · Owner: Lerato Mokoena | $34k | Critical · 84<br>86% High/Crit spend | $29k | Connect Lex Lambda |
| 4 | **payments-core-prod** Critical<br>534439589175 · Payments · Root / Workloads / Production · Owner: Priya Shah | $116k | Low · 20<br>3% High/Crit spend | $23k | DynamoDB Lambda |
| 5 | **digital-channels-prod** Critical<br>978681853473 · Mobile app · Root / Workloads / Production · Owner: Lerato Mokoena | $57k | Moderate · 39<br>25% High/Crit spend | $22k | Cognito AppSync Pinpoint |
| 6 | **core-banking-prod** Critical<br>813958344723 · Core banking · Root / Workloads / Production · Owner: Thabo Dlamini | $122k | Low · 17<br>0% High/Crit spend | $21k | none |
| 7 | **ml-platform-prod**<br>217230843944 · Root / Workloads / Production · Owner: Emma Botha | $47k | Moderate · 39<br>3% High/Crit spend | $18k | Bedrock AgentCore |
| 8 | **fraud-analytics-prod**<br>466806226017 · Root / Workloads / Production · Owner: Emma Botha | $20k | High · 56<br>28% High/Crit spend | $11k | DynamoDB Timestream |
| 9 | **shared-network-prod**<br>263737281768 · Root / Infrastructure | $37k | Low · 18<br>0% High/Crit spend | $6.7k | none |
| 10 | **search-prod**<br>899894212661 · Root / Workloads / Production | $4.0k | Critical · 75<br>70% High/Crit spend | $3.0k | DynamoDB Lambda |
| 11 | **mobile-backend-prod**<br>434423300025 · Root / Workloads / Production · Owner: Lerato Mokoena | $8.0k | Moderate · 31<br>15% High/Crit spend | $2.5k | Lambda |
| 12 | **treasury-reporting-prod**<br>499647302627 · Root / Workloads / Production · Owner: Thabo Dlamini | $2.8k | Critical · 79<br>92% High/Crit spend | $2.2k | DynamoDB QuickSight |
| 13 | **merchant-portal-prod**<br>386906790760 · Root / Workloads / Production | $7.0k | Moderate · 31<br>15% High/Crit spend | $2.2k | Lambda |
| 14 | **doc-store-prod**<br>454519794043 · Root / Workloads / Production | $9.7k | Low · 18<br>0% High/Crit spend | $1.8k | none |
| 15 | **identity-prod** Critical<br>189183446249 · Shared platform · Root / Infrastructure · Owner: Thabo Dlamini | $7.9k | Low · 21<br>0% High/Crit spend | $1.7k | none |
| 16 | **id-verify-prod**<br>267387487076 · Root / Workloads / Production | $8.7k | Low · 18<br>0% High/Crit spend | $1.6k | none |
| 17 | **payroll-api-prod**<br>295250496497 · Root / Workloads / Production | $7.6k | Low · 18<br>0% High/Crit spend | $1.4k | none |
| 18 | **legacy-batch-prod**<br>107537456074 · Root / Workloads / Production | $4.2k | Moderate · 32<br>25% High/Crit spend | $1.3k | SWF Data Pipeline |
| 19 | **branch-queue-prod**<br>717640922470 · Root / Workloads / Production | $7.3k | Low · 18<br>0% High/Crit spend | $1.3k | none |
| 20 | **insurance-quotes-prod**<br>498633168856 · Root / Workloads / Production | $7.2k | Low · 18<br>0% High/Crit spend | $1.3k | none |
| 21 | **disputes-prod**<br>844781801768 · Root / Workloads / Production | $6.9k | Low · 18<br>0% High/Crit spend | $1.2k | none |
| 22 | **investments-prod**<br>514740085648 · Root / Workloads / Production | $8.5k | Low · 14<br>0% High/Crit spend | $1.2k | none |
| 23 | **card-controls-prod**<br>170174047247 · Root / Workloads / Production | $6.4k | Low · 18<br>0% High/Crit spend | $1.2k | none |
| 24 | **pricing-prod**<br>322790187289 · Root / Workloads / Production | $6.1k | Low · 18<br>0% High/Crit spend | $1.1k | none |
| 25 | **reporting-hub-prod**<br>244219422241 · Root / Workloads / Production · Owner: Emma Botha | $7.1k | Low · 14<br>0% High/Crit spend | $997 | none |

Top 25 of 40 accounts by spend at risk. The JSON export lists every account.

## Who owns the risk from the accounts' owner and executive tags

Chris Ndlovu carries the most spend at risk: $109k a month across 4 accounts.

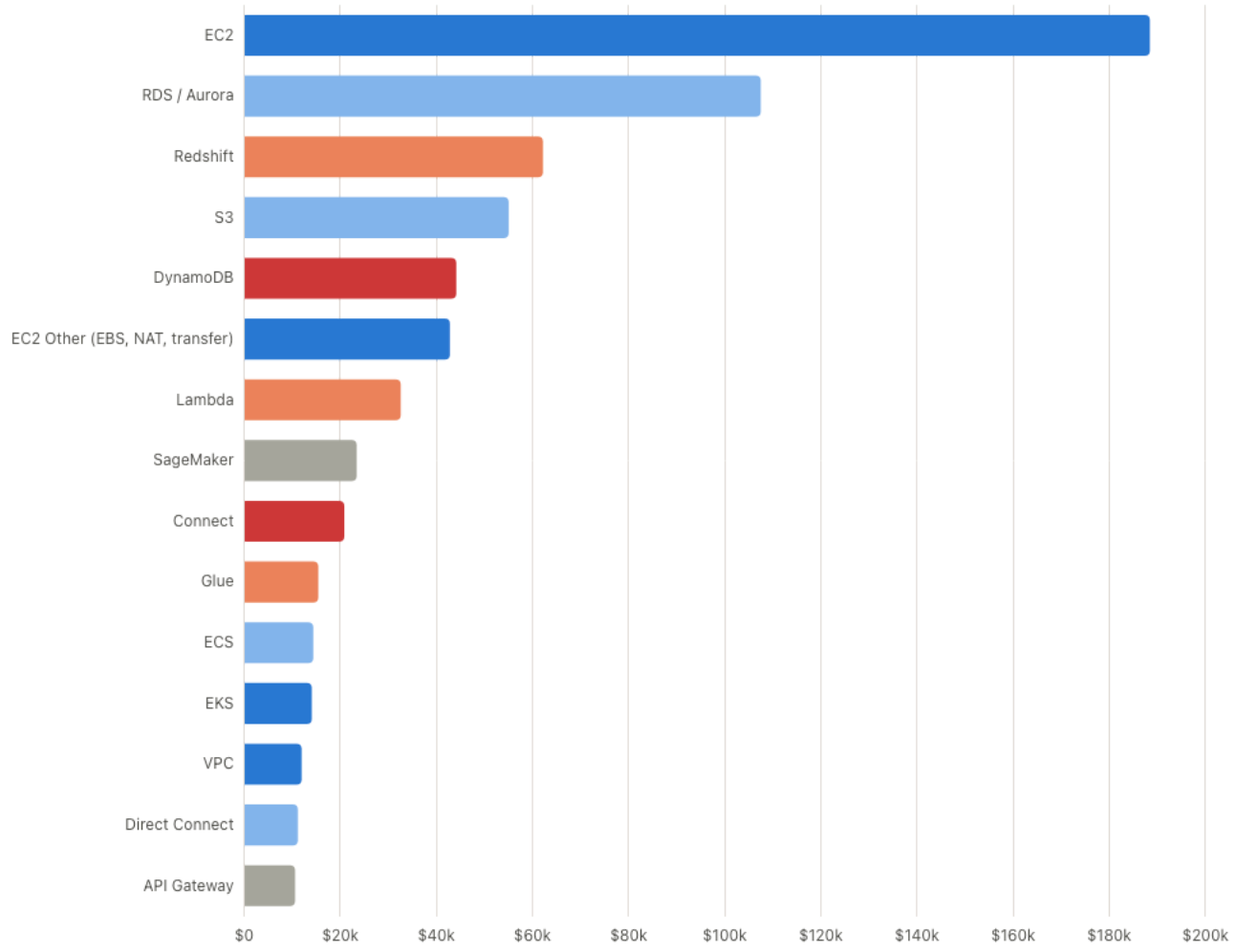
| Executive or owner | Accounts | Per month | Spend at risk / month | High/Critical accounts | Account owners |
|---|---|---|---|---|---|
| **Chris Ndlovu** | 4 | $207k | $109k | 2 | Emma Botha |
| **Alex Morgan** | 2 | $216k | $97k | 1 | Priya Shah |
| **Sam Patel** | 3 | $99k | $53k | 1 | Lerato Mokoena |
| **Jordan Lee** | 3 | $133k | $25k | 1 | Thabo Dlamini |

12 accounts of 40 with spend carry an owner or executive tag. 28 accounts ($172k a month) have none and are left out of this table.

## 4 Services driving lock-in

### Top services by spend

Coloured by exit-risk tier.



## High & critical services in use

| Service | Tier | Per month | Share | Accounts | Change |
|---|---|---|---|---|---|
| **Redshift**<br>Analytics | 4 · High | $62k | 7.5% | 1 | +$23k<br>+45% |
| **DynamoDB**<br>Database | 5 · Critical | $44k | 5.3% | 6 | +$13k<br>+35% |
| **Lambda**<br>Compute | 4 · High | $33k | 3.9% | 10 | +$9.8k<br>+35% |
| **Connect**<br>Business apps | 5 · Critical | $21k | 2.5% | 1 | +$6.6k<br>+38% |
| **Glue**<br>Analytics | 4 · High | $15k | 1.9% | 1 | +$5.4k<br>+42% |
| **Kinesis Streams**<br>Analytics | 4 · High | $9.7k | 1.2% | 2 | +$2.8k<br>+34% |
| **Cognito**<br>Security | 4 · High | $8.0k | 1.0% | 2 | +$2.4k<br>+35% |
| **Step Functions**<br>Integration | 4 · High | $7.4k | 0.9% | 2 | +$1.8k<br>+27% |
| **QuickSight**<br>Analytics | 4 · High | $5.9k | 0.7% | 2 | +$2.0k<br>+41% |
| **AppSync**<br>Integration | 4 · High | $4.3k | 0.5% | 1 | +$1.3k<br>+37% |
| **Lex**<br>AI/ML | 4 · High | $3.1k | 0.4% | 1 | +$1.1k<br>+44% |
| **Pinpoint**<br>Business apps | 4 · High | $2.8k | 0.3% | 1 | +$837<br>+34% |
| **Timestream**<br>Database | 4 · High | $2.0k | 0.2% | 1 | +$783<br>+48% |
| **EventBridge**<br>Integration | 4 · High | $1.7k | 0.2% | 1 | +$510<br>+36% |
| **Amplify**<br>DevOps | 4 · High | $1.3k | 0.2% | 1 | +$382<br>+33% |
| **Bedrock AgentCore**<br>AI/ML | 4 · High | $1.2k | 0.1% | 1 | +$473<br>+47% |
| **SWF**<br>Integration | 5 · Critical | $712 | 0.1% | 1 | +$215<br>+35% |
| **Lake Formation**<br>Analytics | 4 · High | $640 | 0.1% | 1 | +$219<br>+42% |
| **Data Pipeline**<br>Analytics | 4 · High | $327 | 0.0% | 1 | +$86<br>+30% |

## Third-party products billed through AWS

Marketplace software and third-party models on Bedrock. Usually portable: the vendor sells direct or on other clouds, but each contract needs re-papering.

| Product | Billed as | Per month | Accounts | Change |
|---|---|---|---|---|
| Claude Sonnet 4.6 | Bedrock | $3.9k | 1 | +$4.9k |
| Example Streaming Cloud - Pay As You Go | Marketplace (3rd party) | $3.5k | 1 | +$1.6k |
| Example Observability Platform | Marketplace (3rd party) | $2.7k | 1 | +$2.0k |
| Example Data Integration Cloud | Marketplace (3rd party) | $1.7k | 1 | +$564 |
| Example AI Assistant for Enterprise | Marketplace (3rd party) | $1.2k | 1 | +$3.0k |
| Claude Haiku 4.5 | Bedrock | $1.1k | 1 | +$963 |

## 5 Exit cost & time estimate

Total exit cost (indicative)

**$7m to $16m** PROVISIONAL

70 to 180% of a year's AWS spend · about 75% engineering, the rest parallel running

Exit window (time to leave AWS)

**2.5 to 5 years** PROVISIONAL

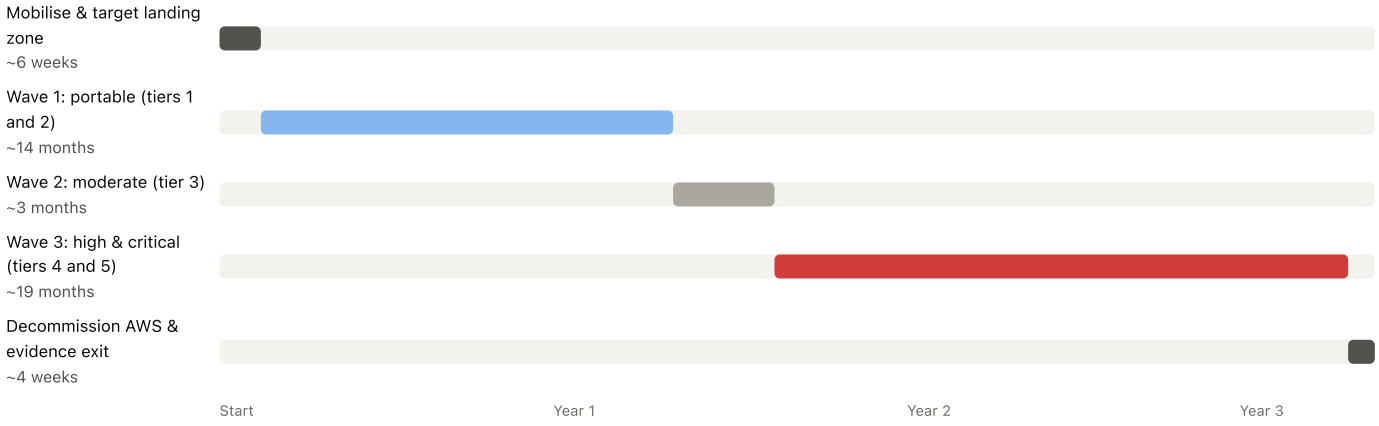
bound by team capacity

Engineering effort

**25 to 60 person-years** PROVISIONAL

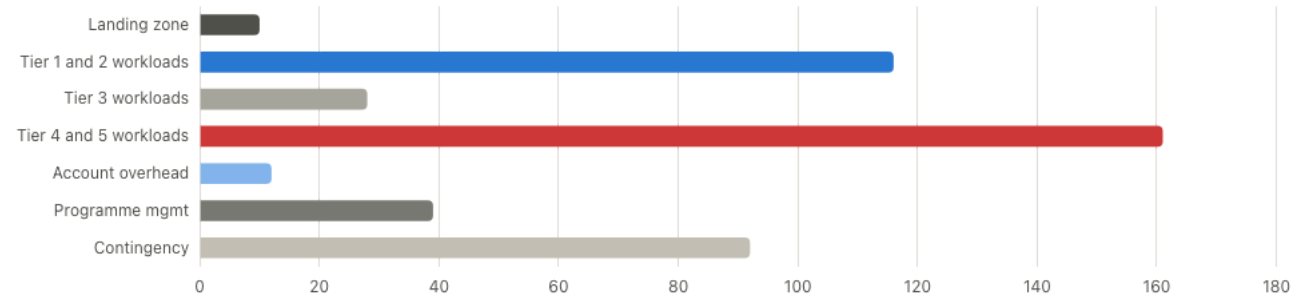
40 material accounts

### Indicative timeline




### Where the effort goes

Person-months, after the delivery-maturity adjustment.




### Your answers and the assumptions behind the estimate

| | | | |
|---|---|---|---|
| Standard CI/CD pipeline<br>Several, partially standard pipelines | Terraform / IaC<br>Terraform with significant drift | Core cloud team<br>10 people at 50% | Application engineers<br>10 FTE |
| Target<br>Microsoft Azure | Blended day rate<br>750 USD | Contingency<br>25% | Parallel running<br>6 months |
| Regulated<br>Yes | AI coding agents<br>Not used | Left out<br>Nothing | |

- Service effort: catalog engineer-weeks per workload × √(monthly spend / 1,000 USD), clamped to 0.25 to 8.0×; resource counts from the deep scan raise it where higher.
- Services under 25 USD/month are scored but not estimated (immaterial).
- 1 engineer-week per material account for network, IAM and decommissioning.
- Target landing zone: 24 weeks (4 + 12 × √(spend / 100,000 USD), max 24) + 0.5 per material account.
- Delivery factor 1.26 = CI/CD 1.0 × IaC 1.1 × target 1.0 × regulated 1.15.
- Programme management and assurance 12%; contingency 25%.
- Engineers are 80% productive on migration work; duration is the longer of effort ÷ capacity and the critical path of the hardest tier present, plus decommissioning.
- Parallel running: 6 months at 50% of current AWS spend (target ramps up as AWS ramps down).
- Excluded: target-cloud run cost after exit, licence changes, AWS commitments still owed after the exit (EDP, Savings Plans, RIs; shown under Timing), and data-transfer-out fees (since March 2024 AWS credits egress for customers leaving AWS, on request to AWS Support and subject to approval, for a move completed within about 60 days; budget egress for any data moved outside that window).
- Figures are quoted as ranges, widened to round numbers: −25% / +50% on cost and effort, −20% / +40% on duration (early-stage estimation cone). The model's single figures are planning midpoints, not forecasts.

## Timing: your deadline and AWS commitments

| Notice period | EDP ends | EDP annual commit |
|---|---|---|
| 24 months | Jun 2028 | $8.0m |

Exit finishes

**Jun 2029 to May 2031** PROVISIONAL

2.5 to 5 years if it starts now

Deadline

**Does not fit** PROVISIONAL

24-month notice

Commitments

**$199k/mo**

$4.2m still to run · last ends Sept 2029

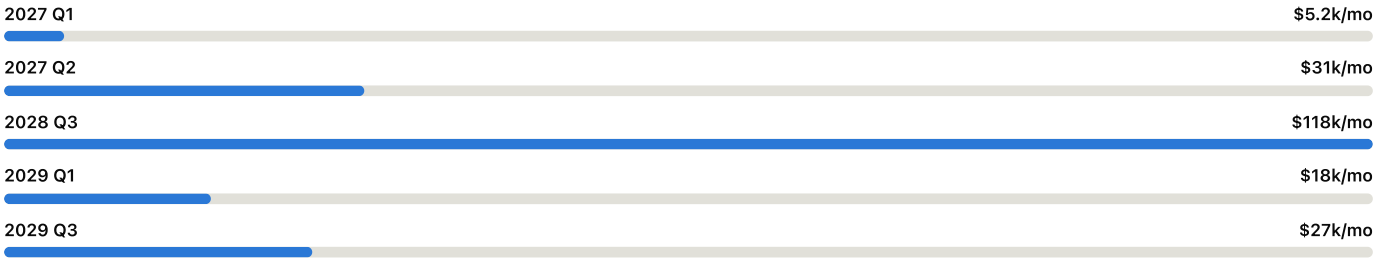
EDP

**ends Jun 2028**

$13.6m of commit left

- The exit does not fit the 24-month notice period: even the fast end runs over, and the slow end is 31 months over.
- About $10k to $100k of commitments would still be owed after the exit finishes; $27k/mo runs past the fastest finish.
- 3-year terms already outlast the fastest exit; 1-year terms stay safe until Jun 2028.
- The EDP ends Jun 2028, before the exit finishes: renew on a shorter term or without a commit.

Monthly commitment that runs off, by quarter




### Savings Plans and reservations

2 Savings Plans and 3 reservations from Cost Explorer, amortised per month. Stranded is what would still be owed between the exit finishing and the commitment ending.

| Commitment | Term | Ends | Per month | Still to run | Stranded |
|---|---|---|---|---|---|
| **Compute Savings Plan**<br>97% used | 3yr<br>No Upfront | Aug 2028 | $118k | $2.6m | – |
| **RDS reservation**<br>89% used | 3yr<br>All Upfront | Sept 2029 | $27k | $927k | $0 to $95k |
| **Redshift reservation**<br>85% used | 3yr<br>All Upfront | Jan 2029 | $18k | $481k | – |
| **EC2 Instance Savings Plan**<br>93% used | 1yr<br>Partial Upfront | May 2027 | $31k | $217k | – |
| **ElastiCache reservation**<br>81% used | 1yr<br>No Upfront | Feb 2027 | $5.2k | $21k | – |

## 6 Resilience: critical accounts, data and a stressed exit

A planned exit is one question. Regulators also ask what happens to critical services if AWS, or one region, fails, and what could move in weeks rather than years. These figures inform; they do not change the score or the estimate.

Critical accounts
5 accounts tagged

Link speed for moving data
10 Gbps

**Critical accounts**
**5**
49% of spend · $4.8m a year

**Correlated failure**
**High · 90**
90% of critical spend in Europe (London)

**No rapid exit**
**24%**
of critical spend is on AWS-only services

**Data to move**
**2.7 PB**
52 days at 10 Gbps

### Correlated failure: where the critical estate runs

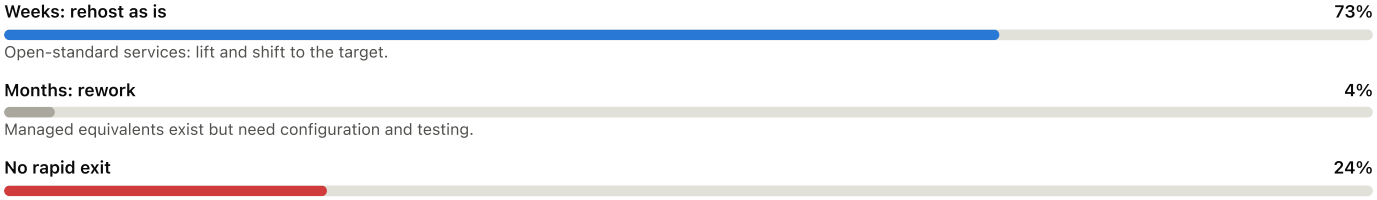
Share of critical spend by region (5 critical accounts). One regional event takes out everything in the top bar at once. A further 3.0% is global services (IAM, Route 53, CloudFront), which share a control plane across regions.




| Channel | Runs mainly in | Isolated from other channels? |
|---|---|---|
| Card payments | Europe (London) 90% | No |
| Core banking | Europe (London) 90% | No |
| Mobile app | Europe (London) 90% | No |
| Payments | Europe (London) 90% | No |
| Shared platform | Europe (London) 90% | No |

### Stressed exit: what could move in a hurry

If AWS stopped serving you with little notice, how fast could critical spend ($4.8m a year) move?




| Critical account with no rapid exit | Per month | AWS-only |
|---|---|---|
| **cards-processing-prod**<br>DynamoDB, Lambda, Kinesis Streams | $101k | 77% |

### Data held: what an exit has to move

Average data stored last month, from Cost Explorer storage usage (GB-months); no data is read. 3.3 PB in all: 2.4 PB live, 261 TB archived and 594 TB of snapshots and backups, which are rebuilt at the target rather than moved.

- Moving 2.7 PB (live and archived) over 10 Gbps takes about 52 days of continuous transfer. That fits the 60-day window of AWS's free data transfer out for customers leaving AWS.
- Without that credit, internet egress at list price would be roughly $140k to $421k.
- Critical accounts hold 890 TB: the data a stressed exit would need first.
- DynamoDB holds 47 TB; the largest account (cards-processing-prod) holds 37 TB. Exports to S3 are not region-limited, but rebuilding in a second region by import from S3 takes 38 import jobs of up to 1 TB each outside us-east-1, us-west-2 and eu-west-1.
- By region: Europe (London) 3.3 PB.

| Service | Live | Archive | Copies |
|---|---|---|---|
| Simple Storage Service | 1.6 PB | 261 TB | – |
| Redshift | 486 TB | – | 121 TB |
| Relational Database Service | 126 TB | – | 262 TB |
| EC2 - Other | 209 TB | – | 167 TB |
| DynamoDB | 47 TB | – | 43 TB |

| Account | Live | Archive | Copies |
|---|---|---|---|
| data-platform-prod | 1.1 PB | 261 TB | 136 TB |
| payments-core-prod Critical | 216 TB | – | 111 TB |
| core-banking-prod Critical | 115 TB | – | 143 TB |
| cards-processing-prod Critical | 163 TB | – | 34 TB |
| ml-platform-prod | 123 TB | – | – |
| digital-channels-prod Critical | 85 TB | – | 24 TB |
| shared-network-prod | 41 TB | – | 33 TB |
| investments-prod | 59 TB | – | 6.6 TB |

## 7 Data quality & scan issues

What this report does *not* cover. Every API error during the scan is recorded here and in the log.

Critical: 0 | Error: 0 | Warning: 2 | Info: 0

| Severity | Area | Account | Issue |
|---|---|---|---|
| Warning | Inventory<br>Ap-Southeast-2 | data-platform-prod | Eventbridge_rules in ap-southeast-2: Rate exceeded `ThrottlingException`<br>→ AWS throttled the request after retries. Re-run, or lower EXITRISK_MAX_WORKERS. |
| Warning | Catalog | – | 'AWS Transform' is not in the service catalog; scored as Moderate (tier 3). $495/month in scope.<br>→ Add it to src/exitrisk/data/service_catalog.yaml (see CONTRIBUTING.md). |

## A Appendix: service exit-risk catalog

Every AWS service is classified into one of five tiers by how much of a workload must change to leave AWS. The tier sets the service's risk score; the recommended path and typical duration are for one workload of that service.

| Tier | Score | Meaning |
|---|---|---|
| 1 · Portable | 10 | Commodity infrastructure or an open standard. Lift-and-shift; config moves, code does not change. |
| 2 · Low | 30 | Managed flavour of an open engine or de-facto standard API. Data and config migrate; little code change. |
| 3 · Moderate | 50 | AWS-specific API with a close equivalent on every major cloud. Code changes at the integration points. |
| 4 · High | 75 | Proprietary programming or data model. The workload must be redesigned, not just re-pointed. |
| 5 · Critical | 95 | No equivalent outside AWS. Rebuild, re-platform or replace with a product; often a business change. |

### Tier 5 · Critical 3 services

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **DynamoDB**<br>Database<br>AI agent help: low<br>$44k/mo in scope | Proprietary key-value/document model. Single-table designs, GSIs, conditional writes, TTL and DynamoDB Streams are baked into application code and data layout. No wire-compatible equivalent outside AWS. | Redesign the data access layer. Common targets are Cosmos DB, Firestore/Bigtable, ScyllaDB (Alternator offers a DynamoDB-compatible API) or PostgreSQL JSONB. Replace Streams consumers. Run dual-writes during cutover. | **Azure:** Cosmos DB<br>**GCP:** Firestore / Bigtable / Spanner<br>**Neutral:** ScyllaDB Alternator / Cassandra / PostgreSQL | 2-6 months per application |
| **Connect**<br>Business apps<br>AI agent help: none<br>$21k/mo in scope | Full contact-centre platform - contact flows, phone numbers, Lex bots, Lambda integrations, recordings and agent desktop. Number porting and agent retraining make it a business transformation. | Select a CCaaS platform (Genesys, NICE, Five9, Dynamics 365 Contact Center), rebuild flows, port numbers, migrate recordings under retention rules, retrain agents. | **Azure:** Dynamics 365 Contact Center<br>**GCP:** Contact Center AI Platform<br>**Neutral:** Genesys Cloud / NICE CXone / Five9 | 6-12 months |
| **SWF**<br>Integration<br>AI agent help: high<br>$712/mo in scope | Legacy proprietary workflow engine with a bespoke decider/worker model; no equivalent. | Rewrite on Temporal (the conceptual successor) or Durable Functions. | **Azure:** Durable Functions<br>**GCP:** Workflows<br>**Neutral:** Temporal | 2-4 months |

### Tier 4 · High 16 services

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **Redshift**<br>Analytics<br>AI agent help: low<br>$62k/mo in scope | PostgreSQL-derived but proprietary MPP warehouse - distribution/sort keys, Spectrum external tables, stored procedures, UNLOAD/COPY from S3, Redshift ML and data sharing are all specific. Downstream BI and ELT jobs depend on its dialect. | Choose a warehouse (Snowflake or Databricks are cloud-neutral; BigQuery / Fabric Warehouse are native). Convert DDL and SQL with a translation tool, re-point ELT (dbt makes this far cheaper), reconcile results in parallel for several reporting cycles. | **Azure:** Fabric Warehouse / Synapse / Snowflake<br>**GCP:** BigQuery<br>**Neutral:** Snowflake / Databricks / ClickHouse | 3-9 months per warehouse |
| **Lambda**<br>Compute<br>AI agent help: high<br>$33k/mo in scope | The function code is portable, but the event model, triggers (S3, SQS, DynamoDB streams, EventBridge), IAM wiring, layers and runtime limits are AWS-specific. Real Lambda estates are mostly glue between other AWS services. | Inventory triggers first. Repackage as containers on Cloud Run / Container Apps / Knative, or rewrite as Azure Functions; replace each event source with the target's equivalent. Migrate with the services it glues together, not alone. | **Azure:** Azure Functions / Container Apps<br>**GCP:** Cloud Run functions<br>**Neutral:** Knative / OpenFaaS on Kubernetes | 4-12 weeks per application |
| **Glue**<br>Analytics<br>AI agent help: high<br>$15k/mo in scope | Spark under the hood, but jobs use GlueContext/DynamicFrame APIs, the Glue Data Catalog, crawlers and Glue workflows. The catalog is usually shared by Athena, EMR, Redshift Spectrum and Lake Formation, so it is a hub. | Port jobs to plain Spark (Databricks, Dataproc, Synapse Spark), migrate the catalog to Unity / Hive Metastore / Dataplex, and replace crawlers with explicit schemas. | **Azure:** Data Factory + Databricks<br>**GCP:** Dataproc / Dataflow + Dataplex<br>**Neutral:** Spark + Airflow + Hive Metastore | 1-4 months |
| **Kinesis Streams**<br>Analytics<br>AI agent help: medium<br>$9.7k/mo in scope | Proprietary streaming API (shards, KCL checkpointing in DynamoDB, enhanced fan-out). Producers and consumers must be rewritten. | Move to Kafka (MSK-equivalent, Confluent, Event Hubs Kafka endpoint) or Pub/Sub; rewrite producers/consumers against the Kafka API to avoid the next lock-in. | **Azure:** Event Hubs (Kafka API)<br>**GCP:** Pub/Sub / Managed Kafka<br>**Neutral:** Apache Kafka / Redpanda | 1-3 months |
| **Cognito**<br>Security<br>AI agent help: medium<br>$8.0k/mo in scope | User pools hold customer identities; password hashes cannot be exported, and triggers / hosted UI are AWS-specific. Customers notice an identity migration. | Move to Entra External ID, Identity Platform, Auth0 or Keycloak using lazy (on-login) migration; plan MFA re-enrolment and app token changes. | **Azure:** Entra External ID<br>**GCP:** Identity Platform<br>**Neutral:** Auth0 / Keycloak / Okta CIC | 2-4 months (customer-facing) |
| **Step Functions**<br>Integration<br>AI agent help: high<br>$7.4k/mo in scope | Workflows are written in Amazon States Language with direct service integrations to 200+ AWS APIs; nothing executes ASL outside AWS. | Rewrite workflows in Logic Apps / Durable Functions, Google Workflows, or Temporal (cloud-neutral and code-first). Redesign each service integration step. | **Azure:** Logic Apps / Durable Functions<br>**GCP:** Workflows<br>**Neutral:** Temporal / Argo Workflows | 1-3 months |
| **QuickSight**<br>Analytics<br>AI agent help: medium<br>$5.9k/mo in scope | Proprietary BI tool; dashboards, SPICE datasets and embedding cannot be exported to other BI products. | Rebuild priority dashboards in Power BI, Looker or Superset; retire the long tail. Treat as a business change, not a technical migration. | **Azure:** Power BI<br>**GCP:** Looker<br>**Neutral:** Apache Superset / Tableau | 1-3 months |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **AppSync**<br>Integration<br>AI agent help: high<br>$4.3k/mo in scope | GraphQL is standard but AppSync resolvers (VTL / APPSYNC_JS), data-source bindings and subscriptions are proprietary. | Re-implement with Apollo Server / Hasura / Azure API Management GraphQL; rewrite resolvers. | **Azure:** API Management (GraphQL) / Hasura<br>**GCP:** Apollo on Cloud Run<br>**Neutral:** Apollo / Hasura | 1-3 months |
| **Lex**<br>AI/ML<br>AI agent help: medium<br>$3.1k/mo in scope | Proprietary bot definitions, intents and Connect integration. | Rebuild bots in Copilot Studio / Dialogflow CX or an LLM-based assistant. | **Azure:** Copilot Studio<br>**GCP:** Dialogflow CX<br>**Neutral:** Rasa / LLM agents | 1-3 months |
| **Pinpoint**<br>Business apps<br>AI agent help: medium<br>$2.8k/mo in scope | Proprietary campaign, journey and segment definitions; sender IDs and registered numbers are tied to AWS. AWS ends support for Pinpoint's engagement features (campaigns, journeys, segments, analytics) on 30 October 2026; SMS, voice and push continue as AWS End User Messaging. | Move to Braze / Twilio / Azure Communication Services, or within AWS to Connect outbound campaigns; re-register sender IDs and short codes. | **Azure:** Azure Communication Services<br>**GCP:** Firebase Cloud Messaging + partner<br>**Neutral:** Braze / Twilio | 1-3 months |
| **Timestream**<br>Database<br>AI agent help: medium<br>$2.0k/mo in scope | Proprietary time-series engine and SQL dialect (Timestream for InfluxDB is the portable variant). | Move to InfluxDB, TimescaleDB, Azure Data Explorer or Bigtable; rewrite queries and ingestion. | **Azure:** Azure Data Explorer<br>**GCP:** Bigtable / BigQuery<br>**Neutral:** InfluxDB / TimescaleDB | 1-3 months |
| **EventBridge**<br>Integration<br>AI agent help: high<br>$1.7k/mo in scope | Proprietary event bus whose value is AWS-native event sources, rules, Pipes and the Scheduler. Event patterns and targets are AWS ARNs. | Re-implement routing on Event Grid / Eventarc or Kafka; replace AWS-native sources with the target's audit/event feeds; move schedules to the target scheduler. | **Azure:** Event Grid<br>**GCP:** Eventarc / Cloud Scheduler<br>**Neutral:** Kafka / Argo Events | 1-2 months |
| **Amplify**<br>DevOps<br>AI agent help: high<br>$1.3k/mo in scope | Opinionated framework that generates Cognito, AppSync, DynamoDB and Lambda backends; front-end hosting is portable, the generated backend is not. | Host front-ends on Static Web Apps / Firebase Hosting / Vercel; re-platform the backend along with Cognito, AppSync and DynamoDB. | **Azure:** Static Web Apps<br>**GCP:** Firebase Hosting<br>**Neutral:** Vercel / Netlify | 1-3 months |
| **Bedrock AgentCore**<br>AI/ML<br>AI agent help: medium<br>$1.2k/mo in scope | Agent runtime, memory, gateway, identity and observability services with AWS-specific APIs. Agent code written with open frameworks is portable; the hosting, memory stores and tool gateway are not. | Keep agent logic in an open framework (LangGraph, Strands, Claude Agent SDK); re-host on Azure AI Foundry Agent Service, Vertex AI Agent Engine or containers; migrate memory to a vector/relational store you control. | **Azure:** Azure AI Foundry Agent Service<br>**GCP:** Vertex AI Agent Engine<br>**Neutral:** Containers + open agent frameworks | 1-2 months |
| **Lake Formation**<br>Analytics<br>AI agent help: medium<br>$640/mo in scope | Proprietary fine-grained permission layer over the Glue catalog; access policies must be re-modelled. | Re-implement as Unity Catalog, Microsoft Purview / OneLake security, or Dataplex policies. | **Azure:** Microsoft Purview / OneLake security<br>**GCP:** Dataplex / BigLake<br>**Neutral:** Unity Catalog / Apache Ranger | 1-2 months |
| **Data Pipeline**<br>Analytics<br>AI agent help: high<br>$327/mo in scope | Legacy proprietary orchestration (in maintenance mode). | Rewrite as Airflow DAGs or Data Factory pipelines. | **Azure:** Data Factory<br>**GCP:** Cloud Composer<br>**Neutral:** Apache Airflow | 1-2 months |

### Tier 3 · Moderate 10 services

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **SageMaker**<br>AI/ML<br>AI agent help: medium<br>$23k/mo in scope | Models (PyTorch, XGBoost, Hugging Face) and containers are portable, but pipelines, feature store, endpoints and Studio domains are SageMaker-specific. | Move training/inference to Azure ML / Vertex AI or Kubeflow/MLflow; rebuild pipelines and feature store. | **Azure:** Azure Machine Learning<br>**GCP:** Vertex AI<br>**Neutral:** Kubeflow / MLflow / KServe | 1-3 months |
| **API Gateway**<br>Integration<br>AI agent help: high<br>$11k/mo in scope | OpenAPI definitions export, but Lambda/VTL integrations, authorizers, usage plans and WebSocket routes are AWS-specific. | Export OpenAPI, re-implement on Azure API Management, Apigee or Kong; rewrite authorizers and mapping templates. | **Azure:** API Management<br>**GCP:** Apigee / API Gateway<br>**Neutral:** Kong / Envoy Gateway | 3-8 weeks |
| **Bedrock**<br>AI/ML<br>AI agent help: medium<br>$7.6k/mo in scope | Foundation models (Anthropic Claude, Llama, Mistral) are available on other clouds and direct from providers, but Bedrock Agents, Knowledge Bases, Guardrails and Flows are proprietary orchestration. | Call the same models via Vertex AI / Azure AI Foundry or the provider's own API; re-implement agents and RAG pipelines with a portable framework; re-run evaluations. | **Azure:** Azure AI Foundry<br>**GCP:** Vertex AI<br>**Neutral:** Provider APIs (Anthropic | 3-8 weeks |
| **Neptune**<br>Database<br>AI agent help: low<br>$6.8k/mo in scope | Graph database using open query languages (Gremlin, openCypher, SPARQL), but loaders, IAM auth and Neptune-specific extensions need rework. | Move to Neo4j, Cosmos DB for Gremlin or JanusGraph; re-validate query behaviour. | **Azure:** Cosmos DB for Apache Gremlin<br>**GCP:** Neo4j AuraDB / Spanner Graph<br>**Neutral:** Neo4j / JanusGraph | 1-3 months |
| **Athena**<br>Analytics<br>AI agent help: high<br>$6.6k/mo in scope | Trino/Presto-based SQL, so queries are largely portable, but it is bound to the Glue Data Catalog, S3 locations and Athena workgroups. | Move tables to Iceberg/Delta, query with Trino, BigQuery or Databricks SQL. | **Azure:** Fabric / Synapse serverless SQL<br>**GCP:** BigQuery<br>**Neutral:** Trino / Starburst | 3-8 weeks |
| **SQS**<br>Integration<br>AI agent help: high<br>$3.6k/mo in scope | Simple semantics, but a proprietary API; every producer and consumer needs a code change. | Move to Service Bus / Pub/Sub / RabbitMQ behind a small messaging abstraction; drain queues at cutover. | **Azure:** Service Bus / Storage Queues<br>**GCP:** Pub/Sub / Cloud Tasks<br>**Neutral:** RabbitMQ / NATS | 2-6 weeks |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **Transcribe**<br>AI/ML<br>AI agent help: medium<br>$3.4k/mo in scope | Proprietary speech API; custom vocabularies need recreating. | Swap to Azure AI Speech / Speech-to-Text or Whisper. | **Azure:** Azure AI Speech<br>**GCP:** Speech-to-Text<br>**Neutral:** Whisper | 2-4 weeks |
| **Textract**<br>AI/ML<br>AI agent help: medium<br>$3.0k/mo in scope | Proprietary document-extraction API and output schema. | Move to Azure Document Intelligence / Document AI; re-map output parsing and re-validate accuracy. | **Azure:** Document Intelligence<br>**GCP:** Document AI<br>**Neutral:** Open-source OCR + LLM extraction | 1-2 months |
| **Firehose**<br>Analytics<br>AI agent help: medium<br>$2.6k/mo in scope | Managed delivery pipeline with AWS-only sources and destinations. | Replace with Kafka Connect, Event Hubs Capture, or Dataflow templates. | **Azure:** Event Hubs Capture / Stream Analytics<br>**GCP:** Dataflow / Pub/Sub subscriptions<br>**Neutral:** Kafka Connect / Vector | 3-6 weeks |
| **SNS**<br>Integration<br>AI agent help: high<br>$1.8k/mo in scope | Proprietary pub/sub and SMS/mobile push API, usually fanned out to SQS and Lambda. | Move topics to Event Grid / Service Bus topics / Pub/Sub; move SMS to a CPaaS provider (Twilio, Azure Communication Services). | **Azure:** Event Grid / Service Bus topics<br>**GCP:** Pub/Sub<br>**Neutral:** NATS / Kafka + Twilio | 2-6 weeks |

**Tier 2 · Low** 18 services

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **RDS / Aurora**<br>Database<br>AI agent help: low<br>$107k/mo in scope | PostgreSQL, MySQL, MariaDB, SQL Server and Oracle engines, plus Aurora (PostgreSQL/MySQL compatible). Schemas and SQL move; Aurora-only features (Global Database, Babelfish, Aurora Serverless scaling, IAM DB auth) and Oracle/SQL Server licensing need attention. | Migrate with native replication or DMS-equivalents (Azure DMS, Database Migration Service) into Azure Database / Cloud SQL / AlloyDB. Cut over per database with a short read-only window. | **Azure:** Azure Database for PostgreSQL/MySQL / Azure SQL<br>**GCP:** Cloud SQL / AlloyDB<br>**Neutral:** Self-managed PostgreSQL (CloudNativePG) | 2-8 weeks per database |
| **S3**<br>Storage<br>AI agent help: low<br>$55k/mo in scope | The S3 API is a de-facto standard implemented by most object stores, but bucket policies, event notifications, lifecycle rules, Object Lock and presigned URL patterns need translating, and large data sets incur egress cost and transfer time. | Copy with rclone / Storage Transfer Service / AzCopy; use S3-compatible endpoints where code depends on the S3 SDK; recreate lifecycle, retention and event rules. | **Azure:** Blob Storage<br>**GCP:** Cloud Storage<br>**Neutral:** MinIO / Ceph RGW | 1-8 weeks (data-volume bound) |
| **ECS**<br>Compute<br>AI agent help: high<br>$14k/mo in scope | Containers are portable but ECS task definitions, service discovery and deployment wiring are AWS-specific and must be rewritten. | Convert task definitions to Kubernetes manifests or Azure Container Apps / Cloud Run definitions. Images move unchanged. | **Azure:** Container Apps / AKS<br>**GCP:** Cloud Run / GKE<br>**Neutral:** Kubernetes / Nomad | 2-6 weeks per service group |
| **Direct Connect**<br>Networking<br>AI agent help: none<br>$11k/mo in scope | Physical circuits; lead times for new carrier connections dominate. | Order ExpressRoute / Cloud Interconnect early (8-16 week lead time is common). | **Azure:** ExpressRoute<br>**GCP:** Cloud Interconnect<br>**Neutral:** Carrier MPLS | 2-4 months (carrier lead time) |
| **EMR**<br>Analytics<br>AI agent help: medium<br>$11k/mo in scope | Open-source Spark, Hive, Presto, Flink. EMRFS, bootstrap actions and step orchestration need adjusting. | Run the same jobs on Databricks, Dataproc or HDInsight; swap s3:// paths and IAM roles. | **Azure:** Databricks / HDInsight<br>**GCP:** Dataproc<br>**Neutral:** Spark on Kubernetes | 1-2 months |
| **Marketplace (3rd party)**<br>Third-party software<br>AI agent help: none<br>$9.2k/mo in scope | Third-party software bought through AWS billing. The product usually exists on other clouds or direct from the vendor; contracts and private offers need re-papering. | Re-contract directly or via the target cloud's marketplace; align with contract renewal dates. | **Azure:** Azure Marketplace<br>**GCP:** Google Cloud Marketplace<br>**Neutral:** Direct vendor contract | Contract-bound |
| **CloudWatch**<br>Management<br>AI agent help: high<br>$8.4k/mo in scope | Metrics, logs, alarms and dashboards are re-creatable, but Embedded Metric Format, Logs Insights queries and alarm-driven automation are AWS-specific. | Standardise on OpenTelemetry and ship to Azure Monitor / Cloud Monitoring / Grafana stack; rebuild alarms and dashboards from code. | **Azure:** Azure Monitor<br>**GCP:** Cloud Monitoring / Logging<br>**Neutral:** Grafana / Prometheus / Loki / Datadog | 2-4 weeks |
| **CloudFront**<br>Networking<br>AI agent help: medium<br>$7.4k/mo in scope | A CDN like any other, but Lambda@Edge / CloudFront Functions and signed-URL schemes are proprietary. | Move to Front Door, Cloud CDN, Cloudflare or Akamai; rewrite edge functions. | **Azure:** Front Door<br>**GCP:** Cloud CDN<br>**Neutral:** Cloudflare / Akamai / Fastly | 1-3 weeks |
| **Network Firewall**<br>Networking<br>AI agent help: medium<br>$5.3k/mo in scope | Suricata-compatible rules; policy structure is AWS-specific. | Translate rules to Azure Firewall / Cloud NGFW or a vendor NGFW. | **Azure:** Azure Firewall<br>**GCP:** Cloud NGFW<br>**Neutral:** Palo Alto / Fortinet | 2-4 weeks |
| **Managed Flink**<br>Analytics<br>AI agent help: high<br>$3.6k/mo in scope | Apache Flink applications are portable; legacy Kinesis SQL applications are not. | Run Flink jobs on Confluent, Ververica, or Flink on Kubernetes; rewrite any legacy SQL apps. | **Azure:** Flink on HDInsight on AKS / Stream Analytics<br>**GCP:** Dataflow / Flink on GKE<br>**Neutral:** Apache Flink | 1-2 months |
| **Backup**<br>Storage<br>AI agent help: low<br>$3.4k/mo in scope | Backup policies and vaults are AWS-specific; the protected data moves with each workload. Retained backups must be kept (or restored and re-protected) to meet retention obligations. | Recreate policies in Azure Backup / Backup and DR; keep the AWS vault read-only until retention expires. | **Azure:** Azure Backup<br>**GCP:** Backup and DR<br>**Neutral:** Veeam / Commvault | Retention-bound tail |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **KMS**<br>Security<br>AI agent help: none<br>$2.4k/mo in scope | Keys cannot be exported from KMS. Data encrypted under KMS must be decrypted and re-encrypted on the target; envelope-encryption code calls the KMS API. | Create keys in Key Vault / Cloud KMS (or an external HSM / BYOK), re-encrypt data during migration, swap the SDK calls. | **Azure:** Key Vault / Managed HSM<br>**GCP:** Cloud KMS / Cloud HSM<br>**Neutral:** HashiCorp Vault / Thales HSM | Moves with the workload |
| **Directory Service**<br>Security<br>AI agent help: medium<br>$2.2k/mo in scope | Managed Microsoft AD - standard Active Directory. | Extend trusts to Entra Domain Services / Managed AD on the target and migrate members. | **Azure:** Entra Domain Services<br>**GCP:** Managed Service for Microsoft AD<br>**Neutral:** Active Directory | 1-2 months |
| **ACM / Private CA**<br>Security<br>AI agent help: low<br>$1.6k/mo in scope | Public certificates are free to reissue elsewhere; a private CA hierarchy must be re-established and trusted by every client. | Reissue public certs on the target; plan a private CA trust rollout (or adopt a cloud-neutral CA such as Vault PKI). | **Azure:** Key Vault certificates<br>**GCP:** Certificate Authority Service<br>**Neutral:** Vault PKI / EJBCA | 2-6 weeks |
| **WAF**<br>Security<br>AI agent help: medium<br>$1.5k/mo in scope | Rule concepts are common; rule syntax and managed rule groups are AWS-specific. | Translate rules to Azure WAF / Cloud Armor / Cloudflare WAF; re-tune in detection mode before enforcing. | **Azure:** Azure WAF<br>**GCP:** Cloud Armor<br>**Neutral:** Cloudflare / Akamai WAF | 2-4 weeks |
| **Global Accelerator**<br>Networking<br>AI agent help: medium<br>$766/mo in scope | Anycast IP front door; static IPs cannot be moved. | Replace with Front Door / global load balancer; plan for client IP allow-list changes. | **Azure:** Front Door / cross-region Load Balancer<br>**GCP:** Global external Load Balancer<br>**Neutral:** Cloudflare Spectrum | 2-4 weeks |
| **Secrets Manager**<br>Security<br>AI agent help: medium<br>$647/mo in scope | Simple secret store, but rotation Lambdas and SDK calls are AWS-specific. | Move secrets to Key Vault / Secret Manager / Vault; rewrite rotation; prefer External Secrets Operator for Kubernetes workloads. | **Azure:** Key Vault<br>**GCP:** Secret Manager<br>**Neutral:** HashiCorp Vault | 1-2 weeks |
| **Batch**<br>Compute<br>AI agent help: high<br>$0/mo in scope | Container jobs on EC2/Fargate; job definitions and queues are AWS-specific. | Move to Azure Batch, Google Batch, or Kubernetes Jobs / Argo Workflows. | **Azure:** Azure Batch<br>**GCP:** Google Batch<br>**Neutral:** Kubernetes Jobs / Slurm | 2-4 weeks |

**Tier 1 · Portable** 12 services

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **EC2**<br>Compute<br>AI agent help: low<br>$188k/mo in scope | Virtual machines running standard operating systems. Images can be exported or rebuilt from configuration management. | Rehost VMs (Azure Migrate, Google Migrate to VMs, or rebuild from Packer/Ansible images). Re-baseline instance sizes on the target. | **Azure:** Virtual Machines<br>**GCP:** Compute Engine<br>**Neutral:** VMware / OpenStack / bare metal | 1-4 weeks per application |
| **EC2 Other (EBS, NAT, transfer)**<br>Compute<br>AI agent help: low<br>$43k/mo in scope | Block storage, snapshots, NAT gateways, elastic IPs and data transfer. Commodity primitives on every cloud. | Recreate volumes and network egress on the target; move data with snapshots or block replication. Budget for one-off egress charges. | **Azure:** Managed Disks / NAT Gateway<br>**GCP:** Persistent Disk / Cloud NAT<br>**Neutral:** SAN / Ceph | Moves with the workload |
| **EKS**<br>Compute<br>AI agent help: medium<br>$14k/mo in scope | Upstream Kubernetes. Manifests and Helm charts run on any conformant cluster; AWS-specific add-ons (IRSA, ALB controller, EBS CSI) are small and swappable. | Stand up AKS/GKE/on-prem Kubernetes, swap cloud add-ons (identity, ingress, storage classes), redeploy via GitOps. | **Azure:** AKS<br>**GCP:** GKE<br>**Neutral:** OpenShift / Rancher / upstream Kubernetes | 2-6 weeks per cluster |
| **VPC**<br>Networking<br>AI agent help: medium<br>$12k/mo in scope | Standard network constructs (subnets, routing, VPN, IPAM); only the console and APIs differ. | Design the target landing-zone network; reuse the IP plan where possible. | **Azure:** Virtual Network<br>**GCP:** VPC<br>**Neutral:** Data-centre networking | Part of landing zone |
| **MSK (Kafka)**<br>Analytics<br>AI agent help: low<br>$7.9k/mo in scope | Standard Apache Kafka. | Mirror topics with MirrorMaker 2 / Cluster Linking and cut consumers over. | **Azure:** Event Hubs (Kafka) / Confluent<br>**GCP:** Managed Service for Apache Kafka<br>**Neutral:** Apache Kafka / Redpanda | 2-6 weeks |
| **FSx**<br>Storage<br>AI agent help: none<br>$5.7k/mo in scope | Managed Windows File Server, Lustre, ONTAP or OpenZFS - all standard file systems. | Replicate to the equivalent managed file system (ONTAP SnapMirror where applicable). | **Azure:** Azure Files / NetApp Files<br>**GCP:** Filestore / NetApp Volumes<br>**Neutral:** NetApp / Windows File Server | 2-4 weeks |
| **OpenSearch**<br>Analytics<br>AI agent help: low<br>$4.7k/mo in scope | Open-source OpenSearch / Elasticsearch APIs (Serverless collections have a reduced API). | Snapshot/restore or reindex into Elastic Cloud, OpenSearch on Kubernetes or Azure AI Search where suitable. | **Azure:** Elastic Cloud on Azure<br>**GCP:** Elastic Cloud on GCP<br>**Neutral:** OpenSearch | 2-4 weeks |
| **ELB**<br>Networking<br>AI agent help: medium<br>$4.2k/mo in scope | Standard L4/L7 load balancing. | Recreate listeners and rules on Application Gateway / Cloud Load Balancing / Envoy. | **Azure:** Load Balancer / Application Gateway<br>**GCP:** Cloud Load Balancing<br>**Neutral:** HAProxy / F5 / Envoy | Moves with the workload |
| **ElastiCache**<br>Database<br>AI agent help: low<br>$4.2k/mo in scope | Managed Redis / Valkey / Memcached - open engines. | Provision the target cache and warm it; caches rarely need data migration. | **Azure:** Azure Cache for Redis<br>**GCP:** Memorystore<br>**Neutral:** Redis / Valkey | Days |

| Service | Why this tier | Recommended migration path | Equivalents | Typical duration |
|---|---|---|---|---|
| **MWAA (Airflow)**<br>Analytics<br>AI agent help: medium<br>$2.0k/mo in scope | Apache Airflow; DAGs are portable apart from AWS operators. | Move DAGs to Cloud Composer, Astronomer or Airflow on Kubernetes; swap AWS operators. | **Azure:** Data Factory Workflow Orchestration Manager / Astronomer<br>**GCP:** Cloud Composer<br>**Neutral:** Apache Airflow | 2-4 weeks |
| **Amazon MQ**<br>Integration<br>AI agent help: low<br>$1.4k/mo in scope | Managed ActiveMQ / RabbitMQ - standard protocols (AMQP, MQTT, STOMP, JMS). | Shovel/federate queues to the target broker and cut clients over. | **Azure:** Service Bus (AMQP) / RabbitMQ on AKS<br>**GCP:** RabbitMQ on GKE<br>**Neutral:** RabbitMQ / ActiveMQ | 1-3 weeks |
| **Route 53**<br>Networking<br>AI agent help: medium<br>$991/mo in scope | DNS zones export as standard zone files. Alias records, health-check routing and Resolver rules need translating. | Export zones, recreate in Azure DNS / Cloud DNS / NS1, then lower TTLs and switch NS records. | **Azure:** Azure DNS<br>**GCP:** Cloud DNS<br>**Neutral:** NS1 / Cloudflare DNS | 1-2 weeks |

### Unclassified services in this estate

Scored as Moderate (tier 3) until added to the catalog.

- AWS Transform: $495/month

## B Appendix: methodology

### Data sources

Spend comes from AWS Cost Explorer in the management (payer) account: amortised cost per account per service per month, excluding tax, credits, refunds and support. Marketplace purchases are grouped as one third-party line. Accounts and OU structure come from AWS Organizations. The optional deep scan counts resources (Lambda functions, DynamoDB tables, state machines, stacks and more) in each member account through a read-only role. All API calls are read-only.

### Scoring

Each service's tier maps to a score: Portable 10, Low 30, Moderate 50, High 75, Critical 95. An account's **inherent risk** is the spend-weighted average of its services' scores.

The report gives **two OU scores**. **By spend** weights each account by its share of in-scope spend, so a small account full of proprietary services scores high on its own yet moves the score very little. **By number of accounts** counts every account with spend equally, so one large account (for example a contact centre on Connect) cannot dominate it. A gap between the two shows whether risk is concentrated in a few big accounts or spread across many. An account's **contribution** is its inherent risk × its weight under the weighting chosen in the bar at the top (by spend unless changed).

Ratings: Low < 25 ≤ Moderate < 40 ≤ Elevated < 55 ≤ High < 70 ≤ Critical.

### Estimate

The cost and time figures depend on the delivery questions under Exit model. Until they are answered, the report marks the figures provisional and keeps them out of the bottom line.

Effort is built bottom-up: each material service in each account contributes the catalog's engineer-weeks for one workload, scaled by the square root of its spend relative to US$1,000/month (and by its resource count where the deep scan ran). Account overhead, a target landing zone, programme management and contingency are added, and the total is adjusted for delivery maturity from the questionnaire. Duration is the longer of effort ÷ available capacity and the critical path of the hardest tier present.

### Limitations

- Spend is a proxy for size; a cheap service can still be hard to move (the deep scan's resource counts partly correct for this).
- RDS spend covers all engines; Aurora-specific features and commercial engines (Oracle, SQL Server) appear in the deep scan's engine breakdown.
- Contractual exit costs (EDP commitments, Savings Plans, Reserved Instances) and target-cloud run costs are outside the estimate.
- The estimate is an order-of-magnitude planning figure for an executive decision, not a delivery quote.